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AI for Bharat Hackathon

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Team Name : npx create

Team Leader Name : Raj Pendkalkar

Problem Statement : AI for Learning & Developer Productivity

Brief about the Idea:

The Problem

Today's AI tools generate code instantly, but they often promote context-less development. New learners and junior developers struggle to understand large codebases, complex architectures, and system flows. As a result, many can build features but cannot confidently explain, modify, or debug the systems they work on. The real challenge today is not writing code but understanding before building.

Our Solution

We are building a Context-Aware Learning and Development AI Agent that transforms how developers interact with technology. Instead of blindly generating code, our system first understands the complete project context, whether it is a new idea or an existing codebase, and converts complexity into structured, simplified, and personalized explanations. It bridges the gap between learning and building by combining system understanding, guided reasoning, and intelligent development assistance in one unified platform.

Who It Is For and How It Helps

- Designed for students, hackathon teams, junior developers, and growing startups
- Analyzes complete codebases, databases, and architectures
- Breaks complex systems into clear, easy-to-understand modules
- Creates a structured learning path based on user knowledge level
- Explains system impact before implementing any feature or change
- Guides development step by step with contextual reasoning
- Encourages real understanding instead of blind AI dependency

how Is It Different from Existing Ideas?

- Tools like Cursor focus on fast in-editor AI assistance, but primarily optimize for speed of implementation.
- GitHub Copilot acts as an intelligent autocomplete system that predicts and generates code snippets.
- Most existing tools respond to prompts but do not deeply enforce architectural understanding or structured onboarding.
- Our solution is context-first and learning-driven, ensuring the developer understands the system architecture, module relationships, and impact of changes before code is generated.
- Instead of only accelerating coding, we accelerate system comprehension and developer growth.

How Will It Solve the Problem?

- Ingests the entire project context, including codebase, dependencies, and workflows
- Converts complex architecture into clear, structured explanations
- Highlights affected modules before any feature implementation
- Guides step-by-step development with reasoning behind decisions
- Reduces onboarding time and builds long-term technical confidence

USP of the Proposed Solution

- First AI platform combining learning engine and development agent
- Architecture-aware and dependency-aware assistance
- Personalized explanations based on user knowledge level
- Promotes skill growth instead of AI dependency
- Designed to make developers smarter, not just faster

- Full Codebase Context Analysis

Reads and understands entire repositories, folder structures, APIs, and databases.

- Architecture Mapping Engine

Automatically generates system architecture and dependency flow explanations.

- Structured Learning Mode

Breaks complex systems into simple modules with guided explanations.

- Personalized Knowledge Level Adjustment

Explains concepts differently for beginners, intermediates, and advanced users.

- Guided Feature Development

Shows impact analysis before implementing new features or modifications.

- Onboarding Assistant for Existing Projects

Acts like an AI mentor for new team members joining a project.

- System Impact Predictor

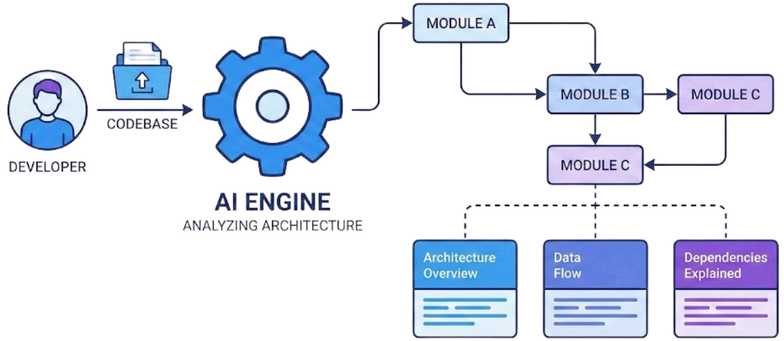
Identifies which modules, APIs, or databases will be affected by changes.

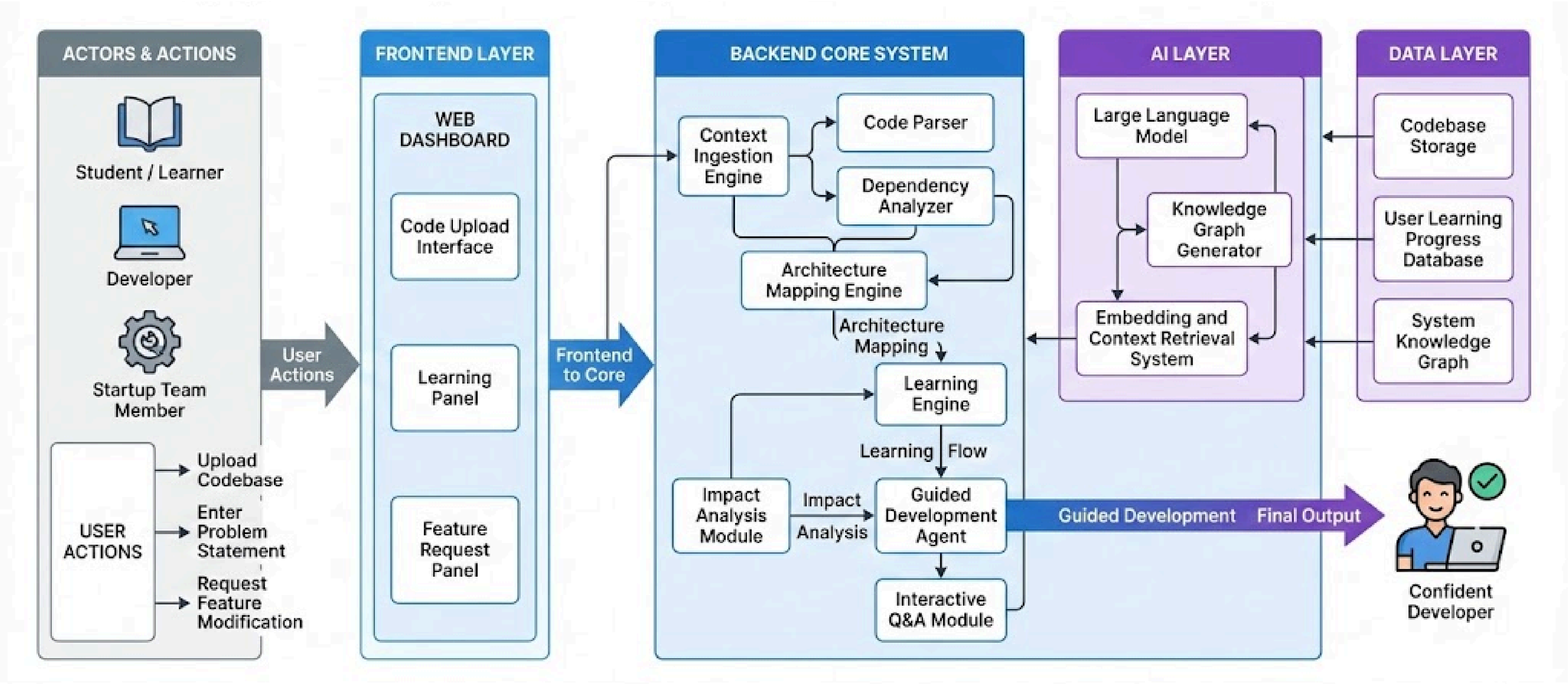
Interactive Q&A Reinforcement

Ensures the user understands before proceeding.

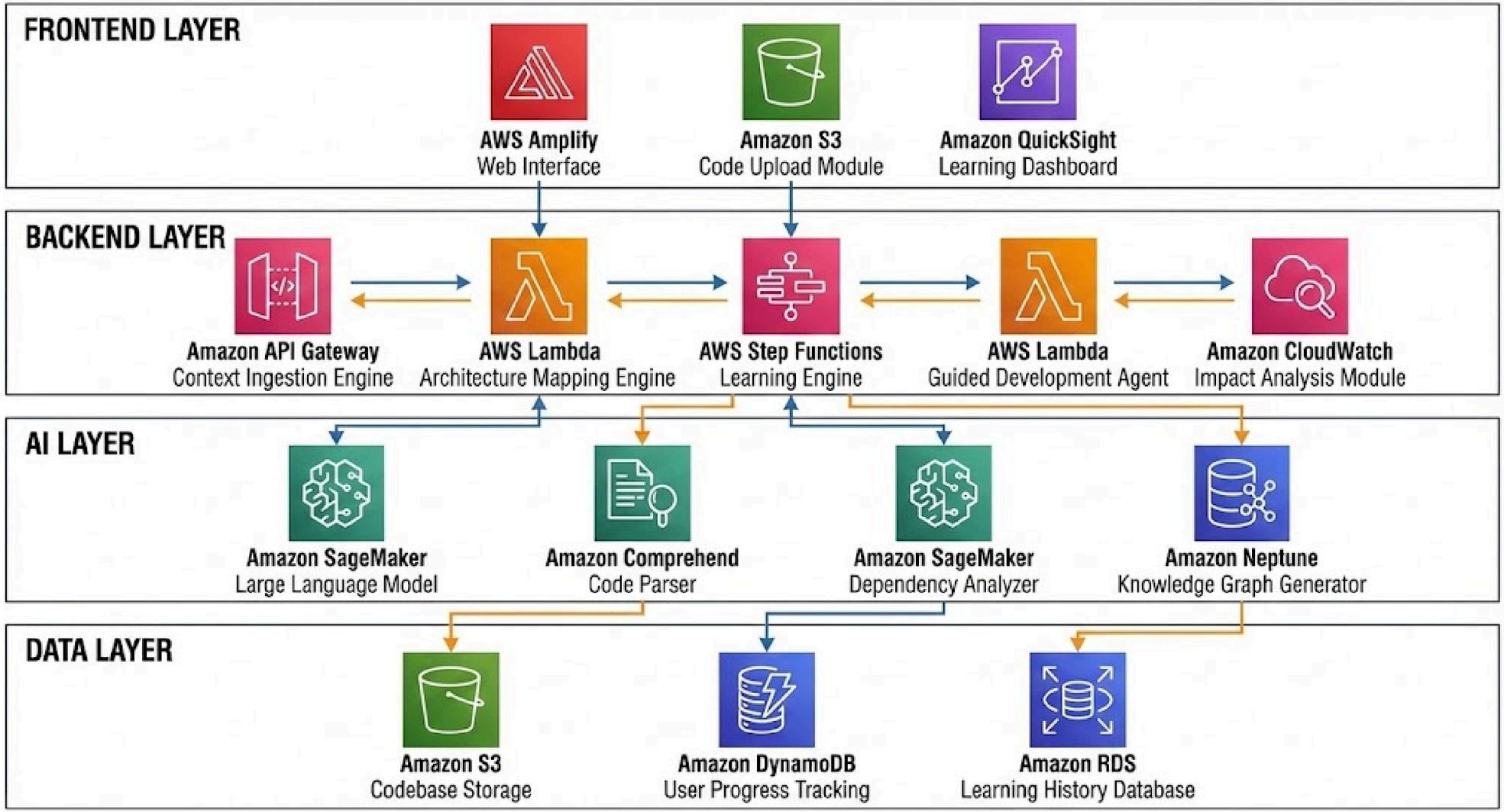
- New Idea Builder Mode

Converts raw problem statements into structured architecture and development plans.




ACTORS & ACTIONS
Student / Learner
Developer
Startup Team Member
USER ACTIONS
Upload Codebase
Enter Problem Statement
Request Feature Modification
User Actions
FRONTEND LAYER
WEB DASHBOARD
Code Upload Interface
Learning Panel
Feature Request Panel
Frontend to Core
BACKEND CORE SYSTEM
Context Ingestion Engine
Code Parser
Dependency Analyzer
Architecture Mapping Engine
Architecture Mapping
Learning Engine
Learning Flow
Impact Analysis Module
Impact Analysis
Guided Development Agent
Interactive Q&A Module
AI LAYER
Large Language Model
Knowledge Graph Generator
Embedding and Context Retrieval System
DATA LAYER
Codebase Storage
User Learning Progress Database
System Knowledge Graph
Guided Development
Final Output
Confident Developer

Architecture



Technologies to be used in the solution:

🖥️ Frontend Layer

- Flutter Web for cross-platform interactive dashboard
- Responsive UI with real-time architecture visualization
- Code viewer and learning panel integration

⚙️ Backend Layer

- Node.js for scalable backend services
- REST APIs for communication between frontend and AI engine
- Repository processing and context management services

🤖 AI and Intelligence Layer

- Large Language Model for reasoning and explanation generation
- Code parsing and static analysis tools
- Embedding-based context retrieval system
- Knowledge graph generation for architecture mapping

🗄️ Data Layer

- Firebase Firestore for user progress and structured data storage
- Vector database for semantic code retrieval
- Secure codebase storage system

🔐 DevOps and Deployment

- Cloud deployment on Google Cloud or similar platform
- Secure authentication system
- Scalable container-based deployment

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Thank You